



REALTEK

Software Development Kit

Switch Product

Release Note

(CONFIDENTIAL: Development Partners Only)

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USING THIS DOCUMENT

This document is intended for use by the system engineer when integrating with Realtek switch products. Though every effort has been made to assure that this document is current and accurate, more information may have become available subsequent to the production of this guide. In that event, please contact your Realtek representative for additional information that may help in the development process.

Table of Contents

SDK Release v2.1.4.53590	4
Summary	4
Changes	4
Bug Fixed	4
Known Issues	7
Delivery List	7

SDK Release v2.1.4.53590

Summary

1. Support RTL8396M-CG.
2. Support RTL8391M-CG.
3. Support RTL839x/5xM-CG.
4. Support RTL838x/3xM-CG.
5. Support RTL8328M/S/L series chips.
6. Support RTL8218B/14FC/18FB-CG.
7. Support RTL8208L.
8. Support RTL8214FB.

Changes

1. Support RTL8396M model.
2. Support RTL8391M model.
3. Update parameters and supported data base for SPI flash chips.
4. RTK APIs add/remove and modify (see the RTK_MS_SDK_API_UserGuide_2.1.4.53590.pdf)
5. Bug Fixed

Bug Fixed

Category	Issue ID	Severity	Description
Boot Loader	R0016545	major	[RTL839x/5xM][U-Boot-2011.12] Support DDR Tx Auto calibration.
Boot Loader	R0016549	major	[Maintenance] [U-Boot-2011.12] Updated DDR and SPI flash support data based.
Boot Loader	R0016494	minor	[RTL839x/5xM,RTL838x/3xM][ENV] Support writing U-Boot default ENV into FLASH function.
Boot Loader	R0016488	minor	[RTL8214F/RTL8214B/RTL8214FB/RTL8212B] Change combo PHY media will fail.
Boot Loader	R0016004	feature	[RTL838xM][RTL8218FB]Update 18FB Parameter to support new board model: RTL8382M_8218B_INTPHY_8218FB.
BSP	R0017715	crash	[RTL838x/3xM][uClibc with NPTL] The system can't boot up to kernel while change to uClibc with NPTL version.
BSP	R0017654	crash	[RTL838x/3xM][Linux 2.6.32.58] Cache alias issue will cause system random Crash.
BSP	R0017317	major	[Linux 2.6.32.58][BSP] Support Kernel

			command line input from U-Boot bootargs.
BSP	R0017465	major	[RTL838x/3xM][BSP][Watchdog] Cannot reboot system when 32M SPI flash is used.
SDK-General	R0017734	block	[RTL8396M] Support RTL8396M 10G model.
SDK-General	R0017706	block	[RTL8391M] Support RTL8391M model.
SDK-General	R0016673	crash	[All CHIPS][SMI] Access a non-initial group SMI device will cause kernel panic.
SDK-General	R0016511	major	[RTL839x/5xM][FlowControl] Change half-duplex back pressure default value to Jam mode.
SDK-General	R0017603	major	[RTL839xM] [Watchdog] SerDes watchdog will access invalid array index.
SDK-General	R0016629	major	[RTL838x/3xM][Port] Add APIs: set phy/get phy, green enable/disable did not used port semaphore to protect.
SDK-General	R0016352	major	[RTL8330M][RTL8212B] 12B phy driver such as setting fiber-1000/100 may failed.
SDK-General	R0016329	major	[RTL838xM] Modify 8X access RTL8231 clock period.
SDK-General	R0017294	minor	[All Chips][DiagShell] Can't enable software control LED.
SDK-General	R0017661	minor	[All Chips][CPU Utilization] The CPU utilization is always in fully loading in IRQ event by "top" app of Linux kernel 2.6.32.58.
SDK-General	R0016484	minor	[RTL839x/5xM,RTL838x/3xM][Diag] Add all Registers and Tables' name and value dump command.
SDK-General	R0016096	minor	[RTL839x/5xM, RTL838x/3xM][SPI Flash] Support SPANSION S25FL256S.
SDK-General	R0017027	minor	[RTL839x/5xM,RTL838x/3xM][General] Lack dispatch API definition in chip mapper configuration.
SDK-General	R0016741	minor	[RTL839x/5xM][RATE] Correct (Tick, Token) default value of Egress Bandwidth.
SDK-General	R0016630	minor	[RTL839x/5xM][RATE] Add 500Mbps to maximum egress bandwidth checking API.
SDK-General	R0016423	minor	[RTL8396M] Correct the configure order of Ingress Bandwidth Control Set Rate.
SDK-General	R0017172	minor	[RTL839x/5xM] [ACL] Get egress tag status action is incorrect.
SDK-General	R0017042	minor	[RTL839x/5xM] [SerDes] Change fiber 1G EEE default state to disable.
SDK-General	R0016821	minor	[RTL839x/835xM] [SerDes] Add diag command to support debug SerDes.
SDK-General	R0017004	minor	[RTL839xM/5xM] [VLAN] Need Port and Port mask parameters for MAC based and IP based VLAN configure APIs.
SDK-General	R0017035	minor	[RTL839x/5xM] NIC driver doesn't translate trap reason completely.
SDK-General	R0017447	minor	[RTL839x/5xM][DAL] Remove PKTDIR_BOTH support in dal_cypress_switch_cpuMaxPktLen_set
SDK-General	R0016208	minor	[RTL839x/5xM] [L2 multicast] L2 multicast display information is incorrect.

SDK-General	R0016428	minor	[RTL839x/5xM] [RTL8218B] SerDes can't link in reset endurance test.
SDK-General	R0016459	minor	[RTL839x/5xM] [ACL] Power off acl block and then power on, the data and mask of entry in the block didn't clear.
SDK-General	R0016487	minor	[RTL839x/5xM] [MDI/MDIX] Port can't link up under endurance test.
SDK-General	R0016064	minor	[RTL839x/5xM][VLAN] Add MAC-based-VLAN with source MAC address mask API.
SDK-General	R0016469	minor	[RTL838x/3xM][QoS] WFQ queue weight not work.
SDK-General	R0016228	minor	[RTL838x/3xM][EEE] By default setting, MAC EEE is enabled, but PHY EEE is disabled, the configuration is inconsistency.
SDK-General	R0016467	minor	[RTL838x/3xM][Diag Shell] Need to add diag-shell command to support dump MIB private counters.
SDK-General	R0015980	minor	[RTL838xM/RTL833xM][WatchDog] Change reset mode from CPU Reset to whole chip reset.
SDK-General	R0016873	minor	[RTL8380M]The internal PHY port is power down when it links in 10M and Rx packet, sometimes MAC still detects port is link up.
SDK-General	R0016546	minor	[RTL8332M][Port] Device port numbers is wrong while target device only connect to an external PHY.
SDK-General	R0015926	minor	[RTL8218B/14FC/18FB][Cross Over] MDI MDIX auto cross-over cannot recover.
SDK-General	R0017200	minor	[RTL8214FC/18FB/14FB][RTL8214B] [DownSpeed] Support UTP DownSpeed 1000M to 100M API.
SDK-General	R0016479	minor	[RTL8214FC/18FB] [PHY ability] 1G ability will be removed of copper mode when set PHY master mode in force fiber mode.
SDK-General	R0016221	minor	[RTL8218B] Correct the PHY probe method.
SDK-General	R0016950	minor	[RTL8218B/14FC/18FB][Cross Over] Support getting MDI/MDIX status command.
SDK-General	R0016790	minor	[RTL8214FC][Fiber] Support Fiber mode Internal Loop-back.
SDK-General	R0016680	minor	[RTL8208D] Improve RTCT accuracy.
SDK-General	R0016660	minor	[RTL8218B] link fail when connect with TestCenter device's PHY force mdi and enable power saving mode.
SDK-General	R0016592	minor	[RTL8214FC/18FB] [API] Add common API in 8218B file.
SDK-General	R0017466	minor	[RTL8218B/8214FC/8218FB] [EEE] 10M EEE capability should not be enabled when EEE configure is enable.
SDK-General	R0017745	minor	[RTL8218FB] [PHY]Can't configure force 1G speed in copper mode.
SDK-UserMode	R0016929	major	[RTL839xM/5xM/8xM/3xM][NIC] System occur unexpected error while NIC descriptor ring size is too large.
SDK-UserMode	R0017713	major	[ALL CHiPs][NIC] Endian check error.
SDK-UserMode	R0016402	major	[RTL838x/3xM][NIC] TX packet will stop



			because of race condition occurs.
SDK-UserMode	R0017756	major	[RTL839x/5xM][NIC] NIC functions are not protected by semaphore.
SDK-UserMode	R0017407	minor	[RTL839x/5xM][NIC] Send Dying gasp packet failed
SDK-UserMode	R0017650	minor	[RTL839x/5xM][OAM]Port malfunction after receive dying gasp and re-enable port.
SDK-UserMode	R0017748	minor	[RTL839x/5xM][OAM CLI] Get/Set dying gasp packet count fail.

Known Issues

Issue ID	Severity	Description

Delivery List

File	Description
rtk-ms-sdk-src-2.1.4.53590.tar.gz	Realtek SDK source code package
rtk-ms-uClinux-src-2.6.19-2.6.32.58-svn53590.tar.gz	Linux kernel reference package
rtk-ms-linux-toolchain-0.9.28.tar.gz	Linux toolchain package for kernel v2.6.19
rtk-ms-uboot-src-1.3.0. 53590.tar.gz	Old boot code package
msdk-4.3.6-mips-EB-2.6.32-0.9.30.3-m32-120424.tar.gz	Linux toolchain package for kernel v2.6.32.58
rtk-ms-uboot-2011.12-src-svn53590.tar.gz	New u-boot 2011.12 package

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